

Connectedness between Derivative Tokens, Conventional Cryptocurrencies and Metals: Evidence from TVP-VAR approach

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ABSTRACT

In recent years, the rise of innovative digital assets has sparked a global investment trend. Investors worldwide are increasingly incorporating these new financial instruments into their portfolios, aiming to optimize returns and capitalize on emerging opportunities in the digital asset market. This strategic shift reflects a growing recognition of the potential value these assets may offer in diversifying and potentially enhancing overall investment performance. The aim of this research is to understand the relationship among derivative tokens, traditional digital currencies, and precious metals. This study utilizes the Time-Varying Parameter Vector Autoregression (TVP-VAR) framework to explore the interconnectedness between derivative tokens, precious metals, and traditional cryptocurrencies. Unlike the Diebold-Yilmaz method, TVP-VAR avoids subjective window size selection, reducing the risk of losing significant observations. This model uniquely computes both static and dynamic connectedness, while also mitigating convergence issues found in other GARCH configurations. The results show that derivative tokens such as Hegic, Switchero, Perpetual, Uma, Palladium and Gold are net return receivers while Injective, Dash, Bitcoin and Litecoin are the net return exporters. Connectedness based on volatility is higher than connectedness based on returns, with the level of connectedness rising significantly during the COVID-19 outbreak, indicating an increase in market activity and fluctuations. Also, economic policy indications and various other indexes are valuable in the analysis of total as well as net directional connectedness. These findings hold significant implications for the digital financial practitioners and investors in cryptocurrency, metals, and derivative markets by offering them valuable

insights to enhance risk management practices and capitalize on the diversification benefits.

KEYWORDS: *Derivative Tokens, Conventional Cryptocurrencies, Metals, TVP-VAR approach.*

SECONDARY KEYWORDS: *Hegic, Switcheo, Perpetual, Uma, Palladium, Gold, Injective, Dash, Bitcoin, Litecoin.*

1. INTRODUCTION

In the wake of the 2008 Financial Crisis, Bitcoin emerged as the first blockchain-based virtual currency which have revolutionized digital finance (Nakamoto, 2008). This groundbreaking development was followed by Willett (2012), "The Second Bitcoin Whitepaper", which introduced the alternative currencies and led to the creation of Master-coin, the predecessor of ICOs (Initial Coin Offerings). ICOs, like IPOs, allow the investors to participate in new crypto launches which mark a significant evolution in the cryptocurrency space. As the crypto market expanded, DeFi emerged as a formidable alternative to traditional finance which has been at the heart of the recent development in the last few years. In DeFi, the primary goal is to replace the institutions of banks and financial players with the latter coming in the form of distributed protocols and smart contracts front-end algorithms which operate on the back of the blockchain technology.

A significant innovation within DeFi is the emergence of derivative tokens, which enable direct peer-to-peer trading and automated contract execution via smart contracts. This increases transparency, liquidity, and reduces the transaction costs. Despite being new, derivative tokens reached a \$3 billion market cap by April 2023. Continuous research is important to understand their relationship with the traditional assets. Studying their impact on cryptocurrencies provides valuable investment insights (Yousaf & Goodwell, 2024). Existing literature, including studies by Yousaf, Gubareva, et al. (2023), Cagli and Mandaci (2023), Umar et al. (2022), Kamal and Hassan (2022), Mo et al. (2022), and Joo and Park (2023), explores the connectivity between crypto markets and real assets like energy tokens, metals, and commodities. Building on this existing research, Yousaf et al. (2024) highlight the relationship of Islamic cryptocurrencies and metal markets. This research aims to extend the analysis by examining the correlations between derivative tokens, traditional cryptocurrencies, and precious metals. Derivative tokens are used for management of risk and speculative pricing, and they are integral to the DeFi landscape which utilizes decentralized networks and smart contracts. For instance, the SNX token on Synthetix facilitates synthetic asset issuance which mimics the real asset returns without direct ownership.

We aim to investigate the novel connections between the traditional commodity markets and these new asset classes in terms of information dispersion. We investigate the risk transmissions between derivative tokens, conventional cryptocurrencies, and precious metals by examining data from December 2020 to February 2023, a period marked by notable price fluctuation in cryptocurrencies. Consequently, the analysis is on the basis of Time-Varying Parameter-VAR (TVP-VAR) framework which is an extension of Diebold and Yilmaz (2014) and it serves as the basis of inter-markets interconnectedness analysis both statically and dynamically. Since the Diebold-Yilmaz method requires the subjective

selection of the dynamic-window size, which could result in the loss of certain important observations and a propensity to display a flat parameter, we prefer the TVP-VAR model. Additionally, TVP-VAR framework has unique benefits which cannot be found in the other frameworks such as VAR-, DCC-, and BEKK-GARCH, especially in testing interconnectedness. The TVP-VAR framework only evaluates the static links between markets of assets, in contrast, alternative frameworks do not. However, it also computes both static and time-rolling connectedness, allowing for the examination of pandemic effects on spillover of return and volatility among the precious metal, derivative, and traditional cryptocurrency markets. Moreover, the TVP-VAR approach has fewer convergence problems than other GARCH configurations (Zhang et al., 2023).

Recent studies on token interconnectivity have expanded from the traditional cryptocurrencies like Bitcoin and Ethereum to include the DeFi assets and their relationship with traditional assets. Milunovich (2022) highlights the security risks in interconnected token networks, while (Yousaf & Yarovaya, 2022) show the correlated risk transmission between traditional assets for instance, NFTs, DeFi assets, and conventional markets, along with herding behavior among the crypto assets. Piñeiro-Chousa et al. (2022) emphasize the role of social media in cryptocurrency movements and suggest that the digital finance tokens can hedge against stock market volatility. Yousaf et al. (2022) link the coexistence of a digital assets and fiat currencies to the COVID-19 pandemic which provide an insights into their complex relationships. Corbet et al. (2023) identify a bubble phenomenon in DeFi and the traditional cryptocurrencies, noting the strong connections between DeFi cryptocurrencies and Ethereum.

Jareño and Yousaf (2023) examines how the COVID-19 crisis affected a relationships among DeFi assets, the stock market, and the metals. Jareño and Yousaf (2023) use the TVP-VAR approach to evaluate interconnectedness of asset tokens by focusing on systemic risk and performance. They investigate the spillover effects on tokens and equities, while (Yousaf, Gubareva, et al., 2023) consider inter-linkages between metal markets, traditional cryptocurrencies, and the NFTs. With limited research on cryptocurrencies and conventional assets, the need for further study is apparent as the investors seek to manage portfolio risks. It is recommended that exposure to precious and industrial metals categories should be altered (Mensi et al., 2021; Mensi et al., 2022) though metals remain valued as the wealth stores and crisis hedges due to their historical success. Our research adds to this literature by examining the connection between derivative tokens, traditional cryptocurrencies, and precious metals by addressing a gap in empirical evidence.

2. DATA AND METHOD

2.1. Data

We use daily return data of the conventional cryptocurrencies, Dash, Litecoin and Bitcoin as well as prominent derivative tokens (Injective-INJ, UMA, PERP Protocol-PERP, Hegic-HEGIC, Carbon Protocol-SWTH) and the precious metals (silver, gold, Platinum and Palladium). The sample is from April 15, 2021, to March 01, 2023. In our study, we have only considered those tokens which have a relatively high market capitalization and for which trading data has been available publicly for more than one and a half years (Yousaf et al., 2022). The *coinmarket.cap* is used for extracting the data of the leading cryptocurrencies and derivative tokens whereas the data of metals is obtained from data stream.

Table 1. *Description of variables*

Tokens	Market capitalization	Description
<i>Injective</i>	\$723,946,103	Injective Protocol with a market capitalization of \$723,946,103 is a decentralized exchange located on the Cosmos Blockchain. It utilizes the INJ token for the purpose of the protocol upgrades, to stake, and as a transaction fee which in turn incentivizes validators and improves the liquidity. Synthetix ensures that holders deposit SNX tokens as collateral to create Synths; therefore, the value of SNX fluctuates with the demand for Synths and staked collateral.
<i>Uma</i>	\$149,075,352	UMA, with a market cap of \$149,075,352, serves as the native currency on the UMA blockchain. It enables the creation, trading, and the transfer of synthetic assets mirroring the real-world assets like gold or stocks, collateralized using UMA tokens. The protocol incentivizes users with rewards and fee discounts for liquidity providers.
<i>Carbon Protocol (SWTH)</i>	\$14,696,973	The purpose of Carbon protocol, a “decentralized cross-chained financial protocol”, is to facilitate the safe and scalable trading of complex financial instruments including bonds, futures, and options.
<i>PERP</i>	\$57,070,792	The main token of the PERP Protocol, a decentralized exchange where users can trade PERP swaps—a particular kind of derivative contract—is the PERP token. A wide range of assets, including cryptocurrency, gold, silver, and others, are supported by this platform. The automated market maker (AMM) mechanism used by the PERP Protocol uses liquidity pools to calculate asset prices and carry out trades.
<i>Hegic</i>	\$12,297,674	Users of the decentralized options trading protocol Hegic can trade European-style options for cryptocurrency. One native token of the Hegic protocol is “HEGIC”. Stable USDC is used to settle Hegic options, while HEGIC tokens are used to pay option premiums. An automatic market maker sets the prices of options on the Hegic

2.2. METHOD

2.2.1. TVP-VAR Model

In the following step of research, there is needed an assessment of the interconnection between the returns and the volatility of the derivative tokens, cryptocurrencies, and precious metals as modeled by the TVP-VAR. This is a model developed initially by Antonakakis and Gabauer (2017) and it is an extension of the earlier model developed by Diebold and Yilmaz (2012) which was further strengthened by the later model from the same scholars namely Diebold and Yilmaz (2014). In this model the variance and covariance matrix are time varying. The TVP-VAR approach has been widely applied in the research which includes those studies by Toparlı et al. (2019), Cao and Xie (2022), and Li et al. (2023). Its advantages over the conventional approaches are notable: it efficiently captures daily processes of government and non-government organizations without the need for arbitrarily selecting a rolling window size, thereby avoiding the smoothing of an erratic parameter estimates. This dynamic adjustment to changes in data preserves valuable information and prevents the loss of observations associated with fixed window sizes. Furthermore, its heteroscedastic approach has been shown to outperform the homoscedastic procedures which is evidenced by Koop and Korobilis (2014)' study. Consequently, this model is well-suited for investigating the interdependence of derivative tokens, conventional cryptocurrencies, and metals, particularly in scenarios with few time series and a low level of data arrivals. The basic TVP-VAR(p) model is as follows:

$$y_t = A_t z_{t-1} + \epsilon_t \quad \epsilon_t | \Omega_{t-1} \sim N(0, \Sigma_t) \quad (1)$$

$$\text{vec}(A_t) = \text{vec}(A_{t-1}) + \epsilon_t \quad \epsilon_t | \Omega_{t-1} \sim N(0, \Xi_t) \quad (2)$$

The *multivariate Kalman filter* is defined as by following Primiceri (2005) and Del Negro and Primiceri (2015) as:

$$\begin{aligned} \text{vec}(A_t) | z_{1:t-1} &\sim N(\text{vec}(A_{t|t-1}), \Sigma_{t|t-1}^A) \\ A_{t|t-1} &= A_{t-1|t-1} \\ \epsilon_t &= y_t - A_{t|t-1} z_{t-1} \\ \Sigma_t &= \kappa_2 \Sigma_{t-1|t-1} + (1 - \kappa_2) \epsilon_t \epsilon_t' \\ \Xi_t &= (1 - \kappa_1^{-1}) \Sigma_{t-1|t-1}^A \\ \Sigma_{t|t-1}^A &= \Sigma_{t-1|t-1}^A + \Xi_t \\ \Sigma_{t|t-1} &= z_{t-1} \Sigma_{t|t-1}^A z_{t-1}' + \Sigma_t \end{aligned}$$

We updated A_t , Σ_t^A , and Σ_t , given the information at time t , by the following steps:

$$\begin{aligned}
\text{vec}(A_t) | z_{1:t-1} &\sim N(\text{vec}(A_{t|t-1}), \Sigma_{t|t-1}^A) \\
A_{t|t-1} &= A_{t-1|t-1} \\
\epsilon_t &= y_t - A_{t|t-1} z_{t-1} \\
\Sigma_t &= \kappa_2 \Sigma_{t-1|t-1} + (1 - \kappa_2) \epsilon_t' \epsilon_t \\
\Xi_t &= (1 - \kappa_1^{-1}) \Sigma_{t-1|t-1}^A \\
\Sigma_{t|t-1}^A &= \Sigma_{t-1|t-1}^A + \Xi_t \\
\Sigma_{t|t-1} &= z_{t-1} \Sigma_{t|t-1}^A z_{t-1}' + \Sigma_t
\end{aligned}$$

Besides, in our estimation we use Kalman gain of “Kt is the Kalman gain which measures how it should be changed in any given state. $\Sigma_{t|t-1}$ is the uncertainty parameter, low (high) parameter indicates it's similar (adjust) to previous state. In addition, Σ_t corresponds to the residual variance trying to express the accuracy of the estimate and the inverse relationship that is, the smaller the value is, the higher the accuracy of the estimate is.” In the study by Diebold & Yilmaz (2014), we metricate generalized connectedness using the time-varying coefficient and covariance matrices. The procedure given below is based on Generalized Impulse Response Functions (GIRF), which resembles the process described by Pesaran and Shin (1998) in their GFEVD analysis. To achieve this, we transform the TVP-VAR model into a vector moving average representation based on the Wold representation theorem, as depicted below:

$$y_t = J'(M_t(z_{t-2} + \eta_{t-1}) + \eta_t) \quad (3)$$

$$= J'(M_t(M_t(z_{t-3} + \eta_{t-2}) + \eta_{t-1}) + \eta_t) \quad (4)$$

$$N_t = K1 \quad \sum_{k=1}^K (v1 \quad \sum_{j=1}^v N_j \quad \phi_{t-k}) \quad (5)$$

with

$$M_t = \begin{pmatrix} A_t & 0_{m(p-1) \times m} \end{pmatrix} \eta_t = \begin{pmatrix} e_t \\ 0 \\ \vdots \\ 0 \end{pmatrix} = J \epsilon_t \quad J = \begin{pmatrix} I \\ 0 \\ \vdots \\ 0 \end{pmatrix} \quad (6)$$

Here, M_t is an $mp \times mp$ dimensional matrix, η_t is an $mp \times 1$ dimensional vector, and J is a $mp \times mp$ dimensional matrix. Limiting k as it approaches the infinity, we obtain:

$$y_t = \lim_{k \rightarrow \infty} J' \left(M_t^{k-1} z_{t-k-1} + \sum_{j=0}^k M_t^j \eta_{t-j} \right) = \sum_{j=0}^{\infty} J' M_t^j \eta_{t-j}, \quad (7)$$

$$y_t = \lim_{k \rightarrow \infty} J' \left(M_t^{k-1} z_{t-k-1} + \sum_{j=0}^k M_t^j \eta_{t-j} \right) = \sum_{j=0}^{\infty} J' M_t^j \eta_{t-j}, \quad (8)$$

$$y_t = \sum_{j=0}^{\infty} B_{jt} \epsilon_{t-j} \quad (9)$$

Where B_{jt} is a matrix of $mp \times m$ dimensions.

This paper employs the generalized variance decomposition method developed by Koop and Korobilis (2014) and Pesaran and Shin (1998). This technique is preferable because the indicated order of variables is not important at this step. In this approach, the H-step-ahead forecast error variance decomposition is represented by the following equation:

$$\phi_{ij}(H) = \frac{\sigma_{jj}^{-1} \sum_{h=0}^{H-1} (e_i' A_h \Sigma e_j)^2}{\sum_{h=0}^{H-1} (e_i' A_h \Sigma A_h' e_j)^2} \quad (10)$$

As the $\phi_{ij,t}(H)$ stands for the unscaled GFEVD ($\sum_{j=1}^K \phi_{ij,t}(H) = 1$), It was suggested by (Diebold and Yilmaz 2009, 2012, 2014) to normalize it by dividing $\phi_{ij,t}(H)$ by the row sums to obtain the value of scaled GFEVD, $\tilde{\phi}_{ij}(H)$

$$\tilde{\phi}_{ij}(H) = \frac{\phi_{ij}(H)}{\sum_{j=1}^N \phi_{ij}(H)} \quad (11)$$

The numerator of Equation (11) measures the total change in a shock for variable (i) while the denominator of the same equation does the same for all shocks in the system. The total connectedness is calculated using the “Generalized Forecast Error Variance Decomposition (GFEVD)” as follows:

$$C_i(H) = \frac{\sum_{i,j=1, i \neq j}^m \tilde{\phi}_{ij,t}(H)}{\sum_{i,j=1}^m \tilde{\phi}_{ij,t}(H)} * 100 = \frac{\sum_{i,j=1, i \neq j}^m \tilde{\phi}_{ij,t}(H)}{m} = 100 \quad (12)$$

As previously seen, the total connectedness of the complete system is also identified in Equation (12) which gives the changes of a shock in one variable as it propagates through the intertwined system.

First, we determine how a shock that changes variable (i) affects a system of variables (j) in step 1. This is known as the system's directional connectedness and is described as follows:

$$C_{i \rightarrow j,t}(H) = \frac{\sum_{j=1, i \neq j}^w \tilde{\phi}_{ji,t}(H)}{\sum_{j=1}^m \tilde{\phi}_{ji,t}(H)} * 100 \quad (13)$$

We also calculate the impact of shock of the variable I received from the system, which consists of “j” variables, after calculating the impact of shock on the system. It is known as the system's directional connection and is described as follows:

$$C_{i \leftarrow j,t}(H) = \frac{\sum_{j=1, i \neq j}^m \phi_{ij,t}(H)}{\sum_{i=1}^m \phi_{ij,t}(H)} * 100 \quad (14)$$

Ultimately, the difference between the impact that is sent to and received from the system is calculated to determine the net effect of variable I on the system.

$$C_{i,t} = C_{i \rightarrow j,t}(H) - C_{i \leftarrow j,t}(H) \quad (15)$$

3. PRELIMINARY ANALYSIS

An overview of the time series structure is given in Fig. 1, with an emphasis on the price dynamics of the derivative tokens under consideration which include Injective, Uma, PERP Protocol, Hegic, and SWTH, as well as metals such as gold, silver, platinum, and palladium, and the conventional cryptocurrencies Dash, Litecoin, and Bitcoin. Initially, all the tokens exhibit an upward trajectory in prices until the onset of 2022 which is followed by a subsequent downward trend. The decline post-2022 can be attributed to several factors that includes the tightening of monetary policy and the collapse of various projects like “Terra, Celsius Network, Voyager Digital, and the FTX exchange”. During the pandemic period, precious metals exhibited a bullish trend, but by December 2022, prices had not fully recovered. The determinants of the price movement due to these events included the interest rate hikes and the fears of unprecedented high inflation in the market, exceeding the popularity of the safe havens. The two main factors contributing to the downward trend in precious metal prices were weak consumer and investment demand but purchases by central banks stepped in to offer some support. The industrial demand dip put a strain on silver prices which were not as affected by this factor compared to Platinum which performed better due to an auto catalyst demand recovery and supply limitations. Notably, gold prices spiked to \$2070 on March 8, 2022, due to the geopolitical tensions and monetary policies which reflects the investors' concerns. The Figure 1 and 2 provide daily price and returns time series for each token, offering valuable insights for investors.

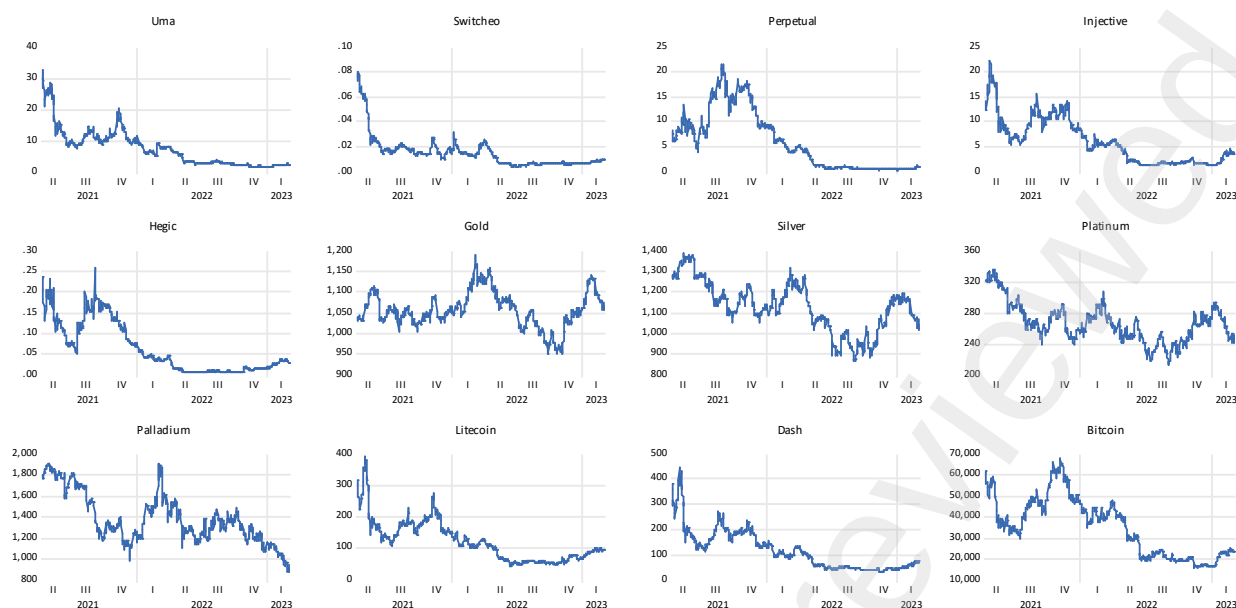


Figure 1: *Trend in derivative tokens, precious metals and leading cryptocurrency*

Figure 2 is a great illustration for the explanation of various returns of the studied 12 assets as presented for different time periods. A visual examination leads to the conclusion of two important observations. The most important finding is that they successfully verified that volatility clustering occurs in derivative tokens, cryptocurrency and precious metals, with several peaks around the post-2021 period which might be resultant of COVID-19 or its spread. Generally, it is seen from metals that they signify comparatively higher number of peaks in the early period of crisis. The research demonstrated that the volatility clustering factor was significant among precious metals, cryptocurrencies, and tokens that are derivatives in nature. This was evidenced by a rising trend was seen beginning from Q1 of 2021 that coincided with the COVID pandemic onset. On the other hand, the base metal and the derivative tokens had more peaks during the initial days of the pandemic.

In general, the research paper demonstrates an increase in the level of volatility, after the spread of Covid-19. Such results may imply a certain level of market predicting power being involved here and perhaps this is clustered volatility which contradicts the random walk theory. The research points out the meaningful volatility clustering among precious metals, derivative tokens, and conventional cryptocurrencies that leads to the crisis stage of the Covid-19 pandemic. The tokens' characteristics are given in table 1.

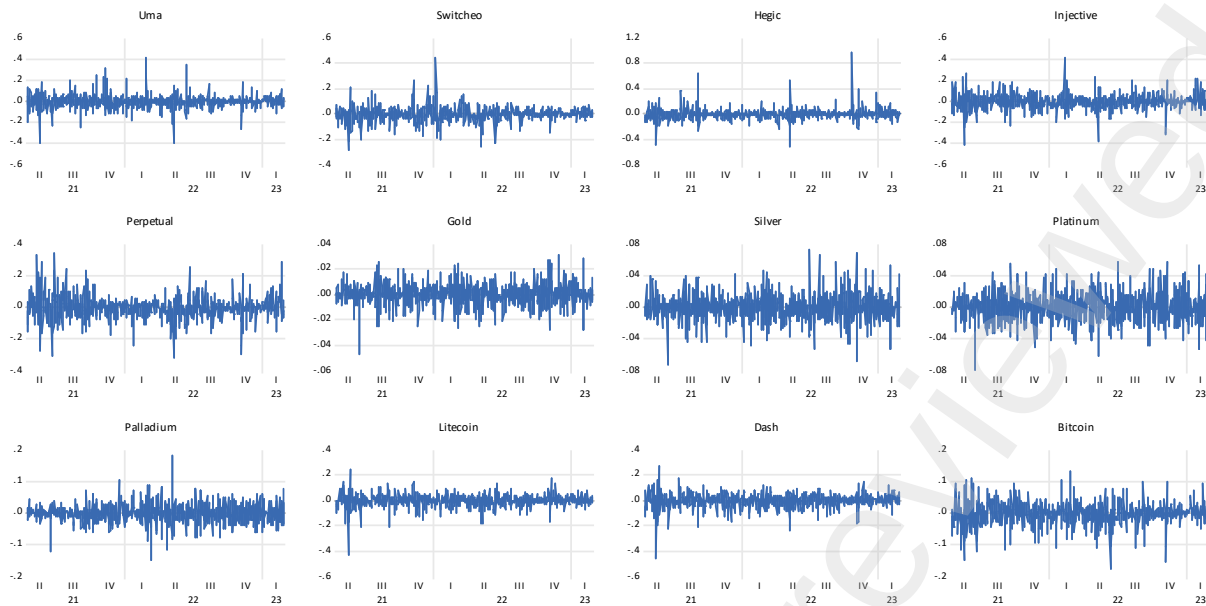


Figure 2: *Returns of derivative tokens, precious metals and leading cryptocurrency*

3.1. Descriptive analysis

Table 2 provides the information about the descriptive statistics for tokens of derivative, metals and cryptocurrencies including statistics for normality and stationarity. The results show that the average return for all types of derivative tokens and currencies is negative with the exception of precious metals such as gold. The Injective, Gold, Dash, Litecoin and the Bitcoin possess negative skewness while some are positive. As seen below, all of the tokens present extreme values of excessive kurtosis, and Jarque-Bera test statistics that are not equal to zero indicating non-normal distributions. ADF-test results reveal that all the assets are stationary in their first differences. There has been a hit on the gains of the derivative tokens and the cryptocurrency in the aftermath of the pandemic.

Table 2. *Descriptive statistics*

Tokens	Minimum	Maximum	Mean	Std.	Skewness	Kurtosis	ADF	JB
				Deviation				
Injective	-0.404	0.404	-0.002	0.074	-.211	4.551	-28.26***	415.938***
Uma	-0.396	0.408	-0.004	0.067	.177	8.190	-29.88***	751.423***
Perpetual	-0.315	0.344	-0.003	0.074	.198	3.622	-26.94***	1797.912***
Hegic	-0.501	0.958	-0.003	0.091	2.443	26.077	-28.22***	1486.458***
Switchco	-0.289	0.448	-0.003	0.059	.225	7.851	-24.58***	298.524***
Gold	-0.048	0.031	0.000	0.009	-.273	1.827	-35.21***	9166.729***
Silver	-0.073	0.072	0.000	0.018	.139	1.824	-34.82***	748.084***
Palladium	-0.146	0.181	-0.001	0.031	.088	2.632	-33.87***	63.820***
Platinum	-0.079	0.057	0.000	0.019	.075	.448	-34.54***	131.023***
Dash	-0.458	0.270	-0.002	0.057	-1.074	7.774	-27.54***	135.052***
Litecoin	-0.428	0.244	-0.002	0.051	-1.095	9.012	-27.32***	15.499***
Bitcoin	-0.176	0.132	-0.001	0.036	-.430	2.850	-26.54***	1797.912***

*** This table shows the daily returns of all variables for the period of April 15, 2021, to March 01, 2023, which are summarized statistically in this table. * Indicates 1% significance, ** indicates 5% significance and *** shows 1% significance.

Note: ADF stands for Augmented Dickey Fuller test and J.B. stands for Jarque Berra test.

Statistics summarized in table 2 illustrate descriptive data for digital coins, precious metals and parameters of normality and stationarity. An average return from all derivative tokens and currencies is witnessing negative returns, while precious metals are exhibiting positive except for palladium. Nevertheless, Injective, Gold, Dash, Litecoin, and Bitcoin are all negatively skewed, but others positively skewed. All the variables have kurtosis levels that are significantly greater than three, where a normal distribution should equal 3. These results are confirmed by the J-BTest results. Such outcomes are verified through ADF stationarity test that proves that all units of asset exhibit stationarity. The pandemic has impacted the returns of all the derivatives and currency as well.

4. EMPIRICAL FINDINGS

At the beginning of this part of the analysis, we summarize our results regarding the static return interconnection between derivative tokens, precious metals, and traditional cryptocurrencies. Then, we move deeper to explore more how these three asset classes interact with each other.

4.1. Static return and volatility spillover

Table 3 shows the time-invariant connectedness between the twelve assets. These encompass digital tokens (Injective, Uma, PERP Protocol, Hegic, and SWTH), metals (gold, silver, platinum, palladium), and other cryptocurrencies (Dash, Litecoin, and Bitcoin). The analysis reveals an interplay between all assets on the network, where the substantial total connectedness index (TCI) illustrates it. TCI is obtained by dividing the sum of external spillovers by the sum of external spillovers and own effects: (769.98/1200). This measure is located at the bottom right side of table 3. The respective overall spillover index (TCI) for the studied period is 64.16%. Furthermore, the diagonal entries are the spillovers among segments (clustering) and the off-diagonal combination refers to the connectedness among assets.

Table 3. Statics returns spillover

	Injective	Uma	Perpetual	Hegic	Switcheroo	Gold	Silver	Palladium	Platinum	Dash	Litecoin	Bitcoin	FRODO
Injective	33.14	9.13	8.33	5.95	5.99	1.1	1.28	1.19	1.95	10.98	10.55	10.4	66.86
Uma	10.25	32.08	6.63	5.44	7.13	1.08	1.68	1.28	1.31	10.81	11.94	10.38	67.92
Perpetual	8.42	7.66	36.17	4.12	6.12	0.79	1.33	2.00	2.29	10	11.16	9.94	63.83

Hegic	10.01	7.12	5.76	39.39	6.06	1.03	0.74	0.92	0.8	9.21	10.96	8.01	60.61
Switcher	6.82	7.84	6.17	4.66	38.71	0.75	1.01	1.02	1.39	10.08	10.77	10.81	61.29
Gold	1.69	0.97	0.82	0.61	0.68	43.19	26.48	8.85	13.98	0.54	1.05	1.13	56.81
Silver	1.47	1.5	1.39	0.45	1.05	24.56	38.7	10.13	17.58	0.72	1.24	1.21	61.3
Palladium	2.17	2.09	2.08	0.93	1.38	8.67	10.71	49.48	17.71	1.29	1.7	1.78	50.52
Platinum	2.45	1.79	1.41	1.08	1.13	14.09	16.17	15.74	41.93	1.16	1.59	1.45	58.07
Dash	8.94	8.59	8.29	5.24	7.82	0.71	1.43	0.99	1.22	26.08	17.49	13.2	73.92
Litecoin	7.92	8.99	7.98	5.92	7.69	0.56	1.53	0.92	1.41	16.67	26.2	14.2	73.8
Bitcoin	8.73	8.39	8.88	4.96	8.91	0.74	1.16	1.01	1.36	14.78	16.15	24.95	75.05
TO	68.86	64.07	57.76	39.37	53.97	54.07	63.54	44.03	60.98	86.24	94.59	82.51	769.98
Inc.Own	102	96.14	93.94	78.76	92.68	97.26	102.24	93.5	102.91	112.32	120.79	107.46	TCI
NET	2	-3.86	-6.06	-	-7.32	-	2.24	-6.5	2.91	12.32	20.79	7.46	64.16

This table reports the static returns spillover between digital assets and energy market during the sample period of April 15, 2021, to March 01, 2023.

The last column of table 3 is known as 'FROM', which is the connectivity that the system forms with all other assets. It measures what each asset receives from other assets and is obtained by summing up the off-diagonal spillover of each row of the matrix. The results indicate that derivatives like the Injective (66.86%), Uma (67.92%), and PERP (63.86%) receive the highest return spillovers from the network. In contrast, the three metals particularly, Palladium (50.52%) along with Gold (56.81%) and Platinum (58.07%) received the least level of spillovers.

The third figure written at the bottom of the column is connectedness value of every metal market against the remaining other asset markets and it is symbolized by 'TO'. The results show that all three cryptocurrencies, namely Litecoin (94.59%), Dash (86.42%), and Bitcoin (82.51%) show the greatest spillover of return transmissions. Conversely, Hegic (39.37%) has the that appears to have the least transmission of the spillovers after Palladium (44.03%) and SWTH (53.97%).

Furthermore, for each of the twelve assets in scope, the directional return connectivity is summarized as "NET" in the final row of table 3. The asset functions as a net beneficiary (exporter) of the net return connectivity from (to) the network, as shown by negative or positive numbers. Therefore, regarding to net spillovers, the derivative tokens such as Uma (-3.86%), Hegic (-21.24%), SWTH (-7.32%), and PERP (-6.06%), and the precious metal commodity Palladium (-6.5%) and Gold (-2.74%) are the net receivers of the return residuals while the remaining precious metals, derivative tokens and all the cryptocurrencies including Dash (12.32%), Litecoin (20.79%), and Bitcoin (7.46%), play the central role of net senders of the return connectedness. In Cagli and Mandaci (2023)

research, they also observed that the natural gas and Palladium markets are correlated, because if Palladium price surges by even number, the price of natural gas surges as well.

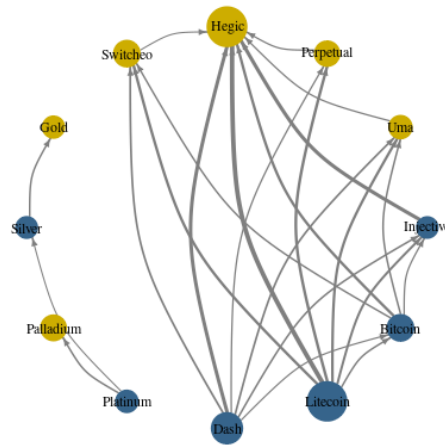


Figure 3: *Flow of Directional return spillover*

The network diagram of the directional return spillovers between twelve assets investigated can be observed in the figure above. In figure 3, we have a network where derivatives' tokens, crypto and metal are connected with regard to returns which are pairwise directed. This expansion builds on the research by Karim et al. (2022) as the direction of the arrows or the spillover in the diagram shows the arrowhead for net receiving and the base for net transmission. Recipient nodes represented by yellow ones and transmitter nodes by blue, symbolize the spillover, where the nodes size determines the strength of the return linkages. This is parallel to the results which are represented in table 3. On the other hand, Hegic, PERP, Uma, SWTH, Palladium, and Gold take the large share of spillover, while Silver, Litecoin, Injective, Platinum, and Bitcoin serve primarily as transmitters in this entire sample period.

Table 4. *Static volatility spillover*

	Injective	Uma	Perpetual	Hegic	Switchero	Gold	Silver	Palladium	Platinum	Dash	Litecoin	Bitcoin	FROM
Injective	18.23	8.54	11.71	5.52	5.3	8.85	10.87	4.72	7.36	6.43	8	4.48	81.77
Uma	7.84	15.19	10.13	8.83	2.94	13.01	4.44	7.69	10.11	8.99	7.04	3.8	84.81
Perpetual	11.31	11.43	17.59	9.59	3.06	11.35	3.4	3.88	11.21	9.38	5.16	2.63	82.41
Hegic	6.37	9.26	10.52	17.32	5.59	8.6	4.03	6.01	8.21	12.06	7.35	4.68	82.68
Switchero	8.4	4.73	5.56	9.08	27.11	4.69	8.62	5.12	4.51	7.59	11.04	3.57	72.89
Gold	8.94	13.96	10.75	8.01	2.06	16.23	3.66	4.95	12.29	11.08	6.99	1.07	83.77
Silver	9.58	5.92	4.2	4.52	6.88	4.62	21.31	12.53	4.66	3.57	9.51	12.68	78.69
Palladium	4.09	10.78	5.21	6.23	3.4	8.65	7.23	19.97	9.11	7.59	8.16	9.58	80.03

Platinum	7.33	10.85	11.04	8.00	2.33	12.75	3.05	6.41	17.6	11.53	7.33	1.79	82.4
Dash	6.87	8.29	8.73	10.67	4.15	9.39	4.83	8.53	9.84	13.79	8.53	6.38	86.21
Litecoin	8.19	6.39	5.62	8.19	6.93	6.69	8.79	10.36	6.96	8.81	16.55	6.52	83.45
Bitcoin	2.25	4.28	2.34	4.5	2.93	2.79	10.1	17.96	3.31	2.45	7.01	40.03	59.97
TO	81.17	94.44	85.79	83.14	45.58	91.37	69.08	88.18	87.57	89.48	86.12	57.18	959.09
Inc.Own	99.4	109.62	103.38	100.46	72.68	107.6	90.39	108.14	105.17	103.26	102.67	97.21	TCI
NET	-0.6	9.62	3.38	0.46	-27.32	7.6	-9.61	8.14	5.17	3.26	2.67	-2.79	79.92

This table reports the static volatility spillover between digital assets and energy market during the sample period of April 15, 2021, to March 01, 2023.

Table 4 presents the static volatility among derivative tokens, precious metals, and conventional cryptocurrencies. The TCI value of static volatility spillover is found to be 50.26% which indicates a stronger volatility connectedness as compared to return connectedness (64.16%). Diagonal elements depict overall market volatility transferral within every market while off-diagonal elements depict directional volatility transferral between individual assets. The "FROM" segment presents volatility movement from the system into constituent elements with Dash (69.14%), Litecoin (68.26%), and Injective (61.78%) contributing the most spillover. Precious metals receive the least volatility spillover, with Palladium (23.59%) having the lowest index from the network. In the "TO" row, Dash (84.10%), Injective (81.07%), and Litecoin (75.27%) exhibit the strongest return spillover effects to the entire system, while SWTH (6.53%) has the least transmission of return spillovers, followed by the Palladium (27.23%). The "NET" row indicates the directional volatility of all assets. Cryptocurrencies like Hegic (-36.36%), SWTH (-45.12%), and Bitcoin (-9.49%) are the net absorbers of volatility spread, while others, including the Injective (19.29%), Uma (7.43%), PERP (7.46%), Palladium (3.64%), Gold (20.71%), Silver (6.26%), Dash (14.96%), and Litecoin (7.01%), transfer the volatility shock.

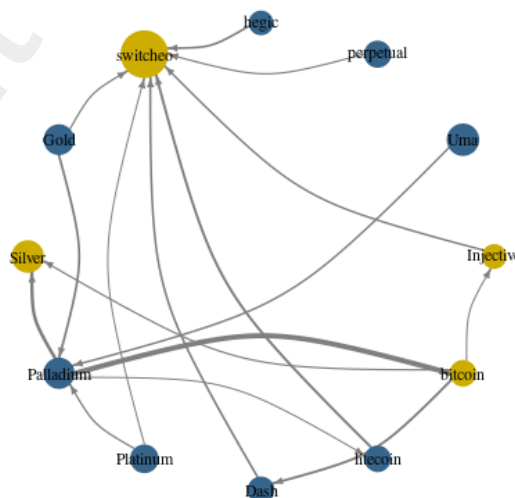


Figure 4: *Flow of directional volatility spillover*

The network diagram of directional volatility spillovers of twelve markets examined can be seen in that figure. Figure 4 shows a network that depicts the interconnectedness of the derivative tokens, metals and conventional cryptocurrencies along static directional volatility, which is represented in the following figure. As per the findings of (Yousaf, Jareño, et al., 2023) and comparable to Figure 3, the orientation of the arrows in the figure indicates the direction of volatility transmission; The arrowhead signifies the net receipt while base denote the net transmission. Hence, nodes in yellow (blue) are disclosed as the recipients (transmitters) of volatility shocks and the size of nodes depicts the degree of pair-wise volatility spillover. The presented data in Figure 4 justifies earlier results which were depicted in table 4. In all, bitcoin, injective, switchero, silver, receive all of the spillover while the Gold, Platinum, Palladium, Litecoin, Perpetual, Uma, Dash, Hegic make up the biggest bulk of the transmitters across the sample timeframe. The volatility spillover features that the assets have could be set to affect portfolio managers and investors.

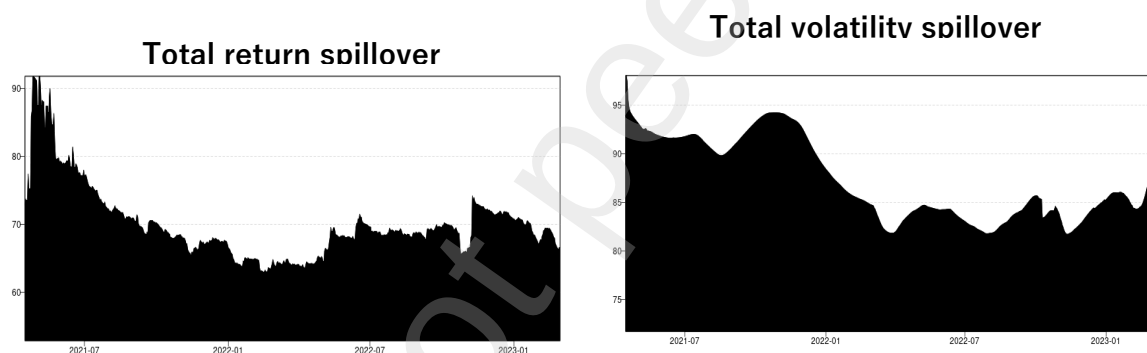


Figure 5: *Total dynamic spillover*

Figure 5 illustrates the dynamics of total return spillovers on the left side and total volatility spillovers presented in right side for the network of twelve study assets.

4.2. Time-varying return and volatility spillover

The study utilizes a dynamic TVP-VAR model to analyze the return and volatility spillovers among derivative tokens, precious metals, and cryptocurrencies. This approach aligns with recent research (Ali et al., 2023; Karim et al., 2022; Yousaf, Gubareva, et al., 2023). Figure 5 displays the rolling trajectory of total portfolio volatility and the connectivity over the entire sample period (see Figure 4) by emphasizing the dynamic nature of spillover phenomena and the need for dynamic portfolio allocations. The spillover effect shown in both the graphs proves the dynamic nature of the whole spillover phenomenon showing that investors should be mindful to keep dynamic portfolio allocations in terms of their

overall strategies. The spillover reduction across the total return value index can be observed in the line chart of total return spillover during the Covid period. Despite a reduction in spillovers during the Covid period, the first quarter of 2021 saw a significant boost coinciding with the pandemic's onset (Ali et al., 2023; Karim et al., 2022; Umar et al., 2021). Spillover effects declined progressively post-recession until the end of 2022, with a resurgence in the first quarter of 2023 possibly due to demand shocks which destabilize the precious metals and other commodities.

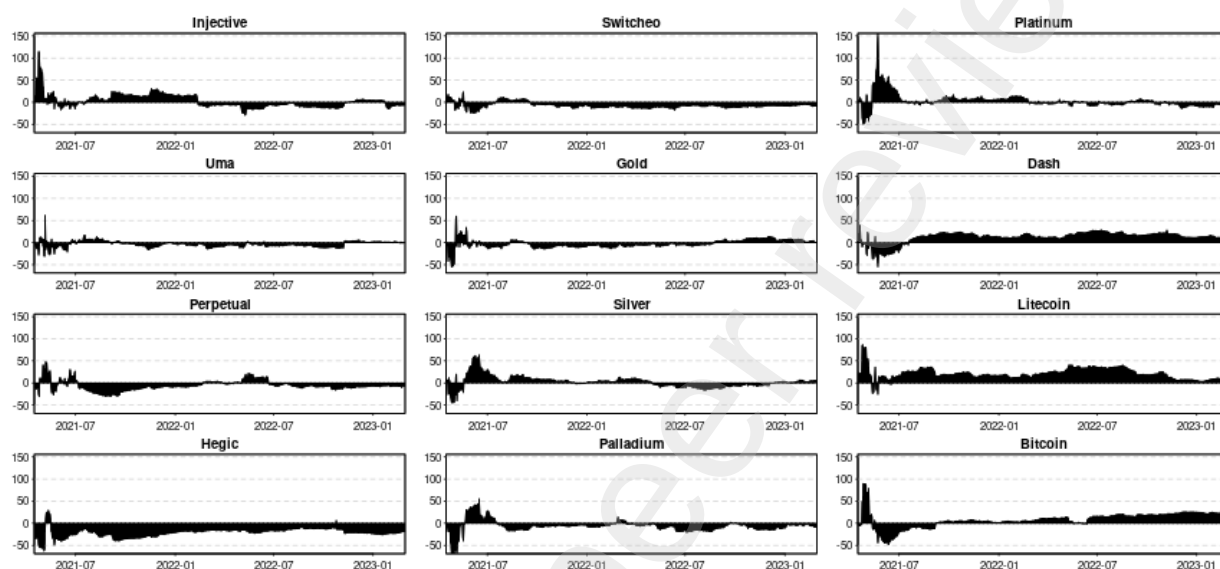


Figure 6: *Net returns spillover derivative tokens, cryptocurrencies and metals*

The above graph reflects net dynamic total return spillover in all 12 markets under investigation.

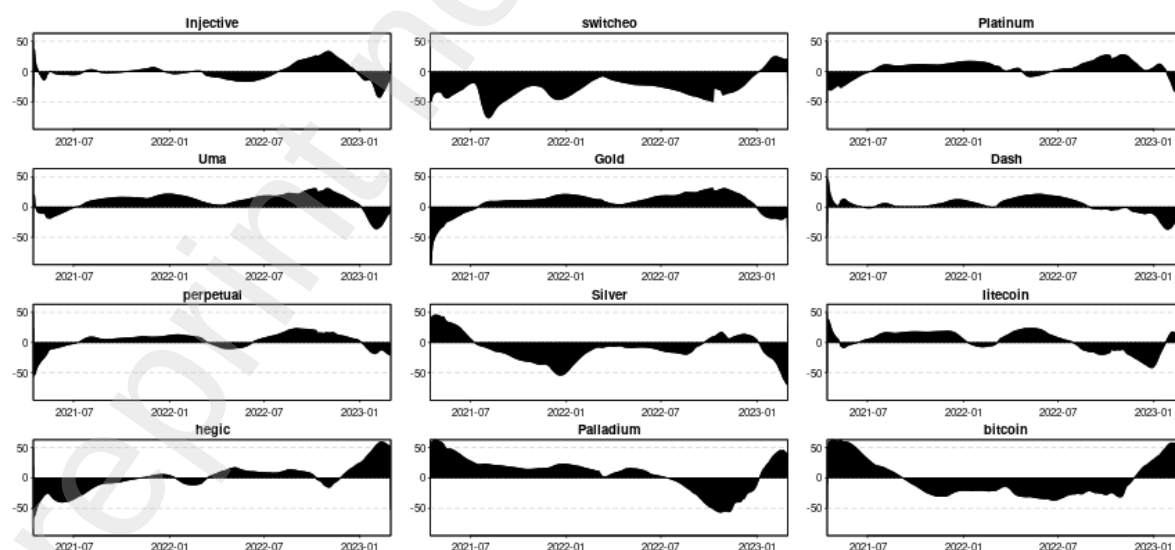


Figure 7: *Net volatility spillover derivative tokens, cryptocurrencies and metals*

The above graph reflects net dynamic total return spillover in all 12 markets under investigation.

In our study, we discuss the findings of time-varying net return and volatility spillovers which are depicted in Figure 6 and 7, respectively. These visualizations reveal the dynamic nature of net connectedness indices over time, showing how assets transition between being receivers and the transmitters throughout the Covid-19 periods. Figure 6 shows that the three derivative tokens consistently act as receivers throughout the sample period - Injective, Perpetual, and Hegic. On the other hand, the transferring role has been emerging for Dash, Litecoin, and Bitcoin coins right after the start of the pandemic and this trend was neutralized in Q1 of 2023. Significant co-movement increases occurred during the pandemic, particularly with assets like Injective, Platinum, Perpetual, and Palladium, indicating modifications in net connection between returns and spillovers during Q1 and Q2 of 2022. Figure 7 presents the volatility connectedness in net terms, where positive values indicate transmitted shocks. The graphics show that the Covid-19 pandemic has either accentuated weaknesses or altered the flows in net terms in major markets by guiding investors and portfolio managers towards flexible strategies for improved portfolio outcomes. Our results align with time-varying fluctuations observed in the energy and related commodities by emphasizing the need for an adaptive approach across fluctuating market dynamics (Shahzad et al., 2021; Umar et al., 2021).

4.3. Determinants of Total Connectedness Index

The dynamic connectedness among the various cryptocurrencies, derivative tokens, and precious metals holds significant implications for portfolio managers, investors, and academic scholars. By comprehending these relationships, stakeholders can mitigate the financial risks and enhance their portfolio returns. To address this issue, we employ a linear regression model to ascertain the net directional and total connectedness index, drawing from the previous studies which have analyzed the connectedness index (Balcilar et al., 2017; Borri, 2019; Ji et al., 2019). Our model integrates macroeconomic factors with an uncertainty indicator to identify their influence on the degree of connectedness across these markets. We thoroughly analyze the variables such as the CBOE Oil Volatility Index (OVIX), Volatility Index (VIX), Energy Index (EI), Euro Currency Volatility Index (ECV), and the Gold Market Volatility Index (GVX), which are pivotal factors shaping the connection dynamics.

For each market segment, we conduct linear regressions to discern the primary factors contributing to the net directional and total connectedness index. By studying the relationship between these variables, we aim to provide the crucial insights to stakeholders into the underlying mechanisms linking the cryptocurrencies, derivative

tokens, and metals and can help them make informed decisions in portfolio management and navigate the challenges of the today's financial markets.

$$TCI_t = \beta_0 + \beta_1 OVX_t + \beta_2 VIX_t + \beta_3 EPU_t + \beta_4 EI_t + \beta_5 GVX_t + \beta_6 ECV_t + \mu t \quad (16)$$

Table 5. *The Determinants of connectedness*

Determinants	Return	Volatility
<i>OVX</i>	5.764*** (0.233)	6.268*** (0.017)
<i>VIX</i>	1.827** (0.076)	2.806** (0.043)
<i>EPU</i>	-5.509*** (0.008)	1.533* (1.115)
<i>EI</i>	-2.641** (0.007)	1.495* (0.004)
<i>GVX</i>	-3.529*** (0.937)	-2.272** (0.080)
<i>ECV</i>	2.642** (0.111)	-7.573*** (0.063)
<i>Constant</i>	7.650*** (31.838)	14.480*** (0.063)
<i>R</i> ²	0.15	0.26

*This table sums up the spillover total determinants results. We have volatility in “gold market predictable (GVX), CBOE volatility index (VIX), economic policy uncertainty (EPU), Oil market volatility index (OVIX), Energy Index (EI), CBOE Euro Currency Volatility Index (ECV)”. The standard error appears in the parenthesis. These symbols **, *, and *** suppose that the coefficient of the corresponding terms is significant at 10%, 5%, and 1% confidence level respectively.*

Following the computation of the return-volatility connectedness index, the focus shifts to determinant analysis, as outlined in (Pham et al., 2022). We select a diverse array of international financial and economic cases to illustrate the framework's linkage. The regression results presented in Table 5 yield intriguing conclusions. Fundamental components such as GVX, EPU, and EI contribute to the total return connectivity index, albeit with negative connections. This suggests that the total connectedness of the system is diminished which indicates a moderating role rather than the direct impacts. Conversely, the US stock market volatility (VIX) and OVX exhibit positive correlations with these components. Furthermore, our analysis reveals the negative impact of GVX and the Euro

Currency Volatility Index (ECV) on volatility connectedness which is consistent with prior research (Adekoya et al., 2021; Mensi et al., 2022; Pham et al., 2022). Additionally, significant differences emerge in the factors affecting the return and volatility which emphasize the importance of recognizing this diversity in impact for investors, policymakers, and the portfolio managers when formulating strategies. Regarding the Energy Index, despite the expectations of a positive correlation between cryptocurrency mining and energy usage due to its energy-intensive nature, our findings present a more nuanced perspective. While the Energy Index exhibits the negative relationship with total net connectedness of volatility, it demonstrates the varying degrees of significance for return volatility, challenging the conventional assumptions and highlighting the complexity of market dynamics.

5. CONCLUSION

Our empirical work contributes to the finance literature by establishing interlinkages between derivative tokens, traditional cryptocurrencies, and precious metals. Using a TVP-VAR model, we explore how returns and the volatility spread across these assets from December 2020 to February 2023, which includes the crisis period. This model accommodates the dynamic financial market linkages without predefined windows by revealing both static and dynamic changes in asset relationships. We find that derivative tokens and cryptocurrencies offer diversification benefits and can be used in metal-based portfolios due to their positive risk-reward relationship. During the COVID-19 period, the extent of spillover increased which indicates higher interconnectivity in the times of crisis. This paper finds that total connectedness index (TCI) values are high, thereby supporting our notion of interconnectedness of assets. Static spillover analysis shows the directional spillover effects shows that the high return spillover on derivative assets like, Injective, Uma, and PERP, while the precious metals experiencing low spillover reception. Static volatility network analysis reveals a high level of relation between the tokens derived from derivatives, precious metals, and cryptocurrencies. The empirical findings of dynamic analysis asserted that the return and volatility spillovers have time-varying characteristics, especially during the pandemic period and, therefore, require dynamic portfolio management. Furthermore, to assess how interconnectivity relates to the various factors of global economic uncertainty, we conducted linear regression analyses on the asset interconnection matrix. Some of the variables consist of GVX which stands for Gold Market Volatility Index, OVIX for Oil Volatility Index, VIX for Stock Volatility Index, EI for Energy Index, EPU for US Economic Policy Uncertainty, and ECV for Euro Currency Volatility Index.

5.1. Limitations and Future Studies

Our study has several limitations that future research can address. Firstly, we focus on a limited selection of the derivative tokens and precious metals. Future work may need to extend this analysis to the additional rare earths and other derived tokens beyond the ones we examined here to provide more diversification outcomes. Secondly, the analysis period, from December 2020 to February 2023, includes the COVID-19 crisis but might not capture the full range of the behaviors of market over different economic cycles. Studies on cyclical patterns and volatility clustering in the market for energy tokens in comparison to other markets might be conducted. Furthermore, even though the TVP-VAR model is reliable for examining time-varying relationships, market dynamics may be too complex for it to fully reflect. Through the application of the advanced methods suggested by Tiwari et al. (2022) and Abakah et al. (2021), such as Markov-switching copulas, we can have a comprehensive understanding of these dynamics. The accuracy and availability of data can also impact the results; future studies should ensure the approach of time and frequency domain quantile statisticians, which was presented by Chatziantoniou et al. (2022), will be used by the researchers so as to make a deeper analysis of the market microstructure no matter the environment. Furthermore, the study does not account for specific regulatory changes, market manipulation, or any other idiosyncratic factors which could affect the relationships between these assets. Future research should include these considerations to refine the analysis.

5.2. Theoretical and Practical Implications

These results offer significant implications and theoretical and practical contributions for investors, portfolio managers, and scholars interested in developing and performing better risk management for portfolios. Theoretically, our study enhances the modern portfolio theory by establishing the interlinkages between derivative tokens, traditional cryptocurrencies, and the precious metals which demonstrate their potential for diversification of portfolio and management of risk. The use of the TVP-VAR model allows for a nuanced understanding of how the returns and volatility spread across these assets over time which reveal the dynamic nature of these relationships, especially during periods of economic uncertainty for instance, COVID-19 crisis. Practically, our results have substantial implications for the investors, portfolio managers, and the policymakers. The positive risk-reward relationship observed among the derivative tokens, cryptocurrencies, and the metals suggests that these assets should be integrated into the portfolio management strategies for diversification and hedging purposes, particularly during the times of crisis.

Author's Contribution:

- Muhammad Adnan: Conceptualization, Formal analysis, Investigation, Project administration, Resources, Supervision, Validation, Writing – review & editing.
- Amir Sohail: Writing – original draft, Formal analysis, Methodology, Validation, Visualization.
- Shrafat Ali Sair: Data curation, Writing – review & editing, Validation, Visualization.
- Alishba Rahman Ullah: Writing – original draft, Project administration, Software, Visualization, Validation.
- Shahzeb Khurshid: Writing – original draft, Formal analysis, Project administration, Validation.
- Volkan Cakir: Supervision, Writing – review & editing, Validation.

Funding Statement: The authors have received no financial support from any private or government sector organization to conduct this research.

Research Ethics Declaration: All authors declare that this manuscript adheres to ethical standards, upholds originality, protects participant confidentiality, avoids conflicts of interest, and prioritizes the safety and wellbeing of all involved. I assume full responsibility for the conduct and presentation of this research.

Conflict of Interest: The authors declare no conflict of interest.

Declaration of generative AI and AI-assisted technologies in the writing process:

While preparing this work, the authors used *Perplexity* and/or *Grammarly* to improve the language quality of the work. After using this tool, the authors reviewed and edited the content as needed and took full responsibility for the publication's content.

Preprint not peer reviewed

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